From Cost Decline to Productivity Transition: How Falling Barriers Accelerate Adoption and Destabilize Growth Regimes

Abstract

Cost barriers at the time of invention predict how fast technologies diffuse. Across 30 technology observations spanning 1605 to 1983, the elasticity of diffusion speed with respect to initial cost barrier is -1.08 ($R^2 = 0.46$, $p < 0.001$). As technologies diffuse unevenly across countries, total factor productivity variance rises before the transition completes, a pattern known in complex systems theory as critical slowing down (CSD). This variance signal appears before technology-specific adoption milestones: it is significant for the telephone ($p = 0.006$) and television ($p = 0.021$), marginal for electricity ($p = 0.063$), and absent for radio. Across structural breaks, electrification-era breaks show rising variance while wars and financial crises do not (Mann-Whitney $p = 0.002$). The pattern replicates before the steam transition in Maddison data extending to 1770 ($p < 0.001$ in three countries). It does not appear before the IT revolution, even at the sector level (Mann-Whitney $p = 0.33$). A higher-frequency test rules out the natural explanation that IT diffused too fast for annual data to resolve: quarterly utilization-adjusted TFP shows no pre-1995 signal either, and simulations calibrated to quarterly noise levels show that even annual sampling would have detected an IT-speed transition 80% of the time. The IT null reflects the scope of the technology, not the frequency of observation: the economy absorbed IT without approaching a critical transition. Operationalized as a causal real-time detector, the signal weakens: backtest AUC is 0.63 with a confidence interval spanning 0.5, and at a fixed false-alarm rate the detector fires before wars more often than before GPT transitions. The signal discriminates populations of breaks in retrospect; it is not yet an operational early-warning instrument.

1 Introduction

Technologies invented later diffuse faster, with adoption lags shrinking substantially over two centuries (Comin and Hobijn, 2010; Comin and Mestieri, 2018). Separately, productivity growth shifts abruptly between regimes, with structural breaks clustering around the arrival of general purpose technologies (Gordon, 2016; Bergeaud et al., 2016). These two facts have been studied in separate literatures. We connect them. Cost barriers at the time of invention determine how fast a technology spreads. Uneven adoption produces rising variance in productivity. The same force that accelerates diffusion generates an early warning signal before the productivity transition arrives.

The mechanism is direct. A technology's initial cost barrier B_j predicts its logistic diffusion rate k_j: the cross-technology regression yields $\log(k) = -3.93 - 1.08 \times \log(B)$, with $R^2 = 0.46$ across 30 technology observations ($N = 30$, covering 24 distinct technologies from 1605 to 1983). Leave-one-out analysis keeps the coefficient between -0.97 and -1.16 with no sign changes. A placebo test that randomly reassigns cost barriers to technologies yields $p = 0.000$ (no permutation out of 500 produces an R^2 as large). As technologies with moderate diffusion speeds spread unevenly across countries, the cross-country variance of TFP rises. In complex systems theory, this is called critical slowing down: a system approaching a bifurcation recovers more slowly from shocks, and its variance grows (Scheffer et al., 2009; Carpenter and Brock, 2006). The variance signal precedes technology-specific adoption inflection points for the telephone ($p = 0.006$) and television ($p = 0.021$) and falls after.

Three further results. Structural breaks associated with electrification show rising pre-break variance (mean Kendall $\tau = +0.23$, $N = 20$), while breaks associated with wars and financial crises show falling variance ($\tau = -0.17$ to -0.36). The difference is significant (Mann-Whitney $p = 0.002$). In Maddison data from the 18th century, variance rises before the steam transition in Britain ($\tau = +0.70$, $p < 0.001$), France ($\tau = +0.83$, $p < 0.001$), and the Netherlands ($\tau = +0.73$, $p < 0.001$). Cross-country dispersion of TFP levels rises in the decades before electrification ($\tau = +0.56$, $p < 0.001$).

Several results fail. IT-era technologies show reversed CSD: the PC (pre $\tau = -0.41$, post $\tau = +0.53$) and mobile phones (pre $\tau = -0.21$, post $\tau = +0.41$) have variance *rising after* the adoption inflection, not before. At the sector level, IT-using sectors do not show stronger CSD than Baumol sectors (Mann-Whitney $p = 0.33$). When the cost barrier is decomposed into three pillars (cognitive, energy, coordination), the overall R^2 rises to 0.63, but the individual pillars are too correlated to be reliably separated. The cross-country panel has the wrong sign. These failures constrain the theory.[1]

The GPT framework of Bresnahan and Trajtenberg (1995) and Helpman (1998) established that general purpose technologies generate complementary innovation and restructuring costs. The diffusion literature, from Griliches (1957) and Mansfield (1961) through Comin and Hobijn (2004) and Comin and Hobijn (2010), has documented cross-country adoption patterns and the role of barriers. CSD was introduced as a generic early warning signal for critical transitions by Scheffer et al. (2009) and Dakos et al. (2008); Carpenter and Brock (2006) formalized rising variance as a leading indicator. Economics applications remain scarce (Diks et al., 2019; Battiston et al., 2016). We apply CSD to productivity regime shifts and connect it to the mechanism of heterogeneous technology diffusion.

Section 2 presents the framework. Section 3 describes the data. Section 4 estimates the cost-speed relationship. Section 5 tests for CSD before adoption inflection points. Section 6 extends the analysis to two centuries of data. Section 7 confronts the IT puzzle. Section 8 asks whether the

[1] Section 7 tests whether the IT null merely reflects diffusion too fast for annual data to resolve, using quarterly utilization-adjusted TFP (Fernald, 2014). It does not: there is no signal at quarterly frequency, and calibrated simulations show annual sampling already had adequate power at IT-speed transitions.

signal supports a real-time detector, and answers honestly: not yet. Section 9 presents robustness checks. Section 10 concludes.

2 Framework

2.1 Cost Barriers and Diffusion Speed

A technology j faces an initial cost barrier B_j: the expense of producing, deploying, and learning to use it, measured relative to prevailing income at the date of invention. We construct B_j as a composite of three pillars: cognitive cost (computation, design, and learning), energy cost (power and fuel), and coordination cost (communication, logistics, institutional setup). Each pillar is measured relative to per capita GDP at the date of invention, so that a technology requiring the same absolute resources in 1950 as in 1800 has a lower barrier because incomes have risen.

Technology j diffuses across countries following a logistic path with rate parameter k_j. The hypothesis is that k_j is determined by B_j:

$$k_j = \frac{\gamma}{B_j}, \qquad \text{or equivalently} \qquad \log(k_j) = \log(\gamma) - \beta \log(B_j), \tag{1}$$

where $\beta = 1$ under the baseline prediction that a doubling of the cost barrier halves the diffusion speed.

A technology requiring large fixed investments, extensive infrastructure, or specialized skills will spread slowly. One requiring little of these will spread quickly. Since all three cost pillars have fallen over historical time, each successive technology faces a lower barrier and diffuses faster.

2.2 Heterogeneous Adoption and Critical Slowing Down

Uneven diffusion creates temporary heterogeneity in production methods. Early adopters reorganize around the new technology; laggards continue with the old. TFP variance across countries rises during the transition.

Consider a state variable x representing a country's productivity, governed by a potential $V(x;\lambda)$ where λ is the cost of the new technology. When λ is large, the equilibrium is stable. As λ falls toward a critical value λ^*, the curvature at equilibrium flattens:

$$V''(x^*;\lambda) \to 0 \qquad \text{as} \quad \lambda \to \lambda^*. \tag{2}$$

The system recovers more slowly from perturbations, and variance grows. This is critical slowing down (Scheffer et al., 2009; Kuehn, 2011). The phenomenon is generic across bifurcation types; the specific form of V does not matter for the qualitative prediction.

In productivity terms, x is TFP and perturbations are the idiosyncratic shocks hitting each country. As the old regime loses stability, a positive shock can tip a country into early adoption while a negative shock keeps it in the old equilibrium. Cross-sectional variance of TFP rises. Once most countries have adopted, the new equilibrium stabilizes and variance falls.

2.3 Predictions

Four predictions follow.

H1: The elasticity of diffusion speed with respect to cost barrier is approximately -1.

H2: TFP variance rises before the adoption inflection point of a diffusing technology and falls after.

H3: CSD appears before productivity breaks associated with GPTs (which involve genuine regime change) but not before breaks caused by exogenous shocks (wars, financial crises) that do not involve technology-driven bifurcation.

3 Data

Technology adoption. Cross-country adoption data come from the Historical Cross-Country Technology Adoption dataset (HCCTA) of Comin and Hobijn (2004), updated through Comin and Hobijn (2010). We use approximately 80 technology variables across 23 OECD countries from the broader HCCTA dataset. We use six technologies with sufficient cross-country coverage for the CSD analysis: electricity, telephone, radio, television, personal computers, and mobile phones. Adoption inflection points are defined as the year each technology first crosses 50% of its maximum adoption level in a given country. These inflection points are computed entirely from the adoption data and are independent of TFP.

Cost barriers. We construct cost barrier indices B_j for 30 technologies spanning 1605 (newspapers) to 1983 (personal computer). Each barrier is a weighted composite of three normalized cost pillars (cognitive, energy, coordination) evaluated at the technology's launch year, plus a distribution-type component reflecting whether the technology requires physical infrastructure or can travel over existing networks. For pre-industrial technologies, we use wage and price data from Allen (2001). Energy costs draw on Fouquet and Pearson (2006) and Fouquet (2008). Computing costs follow Nordhaus (2007). The three-pillar decomposition (cognitive, energy, coordination) uses the same sources. These indices involve judgment calls.[2]

Diffusion speed. Logistic diffusion rates k_j are estimated from the HCCTA data by fitting $S(t) = 1/(1+e^{-k(t-t_0)})$ to each country's normalized adoption path and taking the median k across countries. The BCL productivity dataset (Bergeaud et al., 2016) provides alternative adoption timing for robustness.

[2] For several technologies, the "unit cost at introduction" requires defining what a unit is. For the steam engine, we use the cost of a Newcomen engine circa 1712 relative to British per capita GDP. For electricity, we use the cost per kilowatt-hour of generating capacity in 1882. These choices affect levels but not rankings, and the leave-one-out analysis in Section 4 shows the main result is not driven by any single technology.

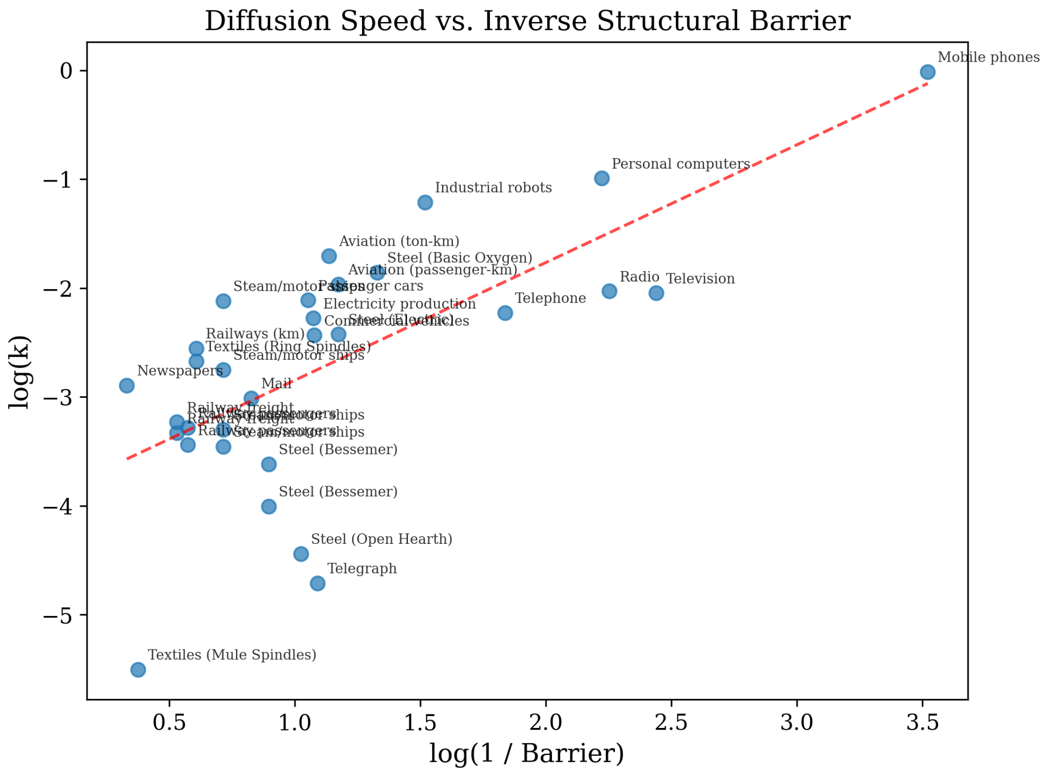

Figure 1: Log diffusion speed vs. log cost barrier for 30 technologies (1605–1983). The fitted line has slope −1.08 ($R^2 = 0.46$). Each point is one technology; the cost barrier is a weighted composite of three normalized cost pillars evaluated at the technology's launch year.

Total factor productivity. Country-level TFP comes from the Long-Term Productivity database of Bergeaud et al. (2016), covering 23 OECD countries from 1890 to 2019. For the long-run analysis, per capita GDP from the Maddison Project Database (Bolt et al., 2018) serves as a proxy for productivity, covering selected countries from 1270 to 2018.

Sector-level data. Sectoral value added and employment come from the Groningen Growth and Development Centre (GGDC) 10-Sector Database, covering up to 42 countries from 1950 to 2012. We classify sectors as IT-using (finance, business services, wholesale/retail) or Baumol (government, education, health) following Baumol (1967) and Jorgenson et al. (2008).

Structural breaks. We detect structural breaks in TFP growth using the Bai and Perron (1998) procedure with a 15% trimming. Breaks are classified into four categories based on timing: electrification (1910–1935), WWII (1936–1948), productivity slowdown (1965–1982), and IT/GFC (1990–2010).

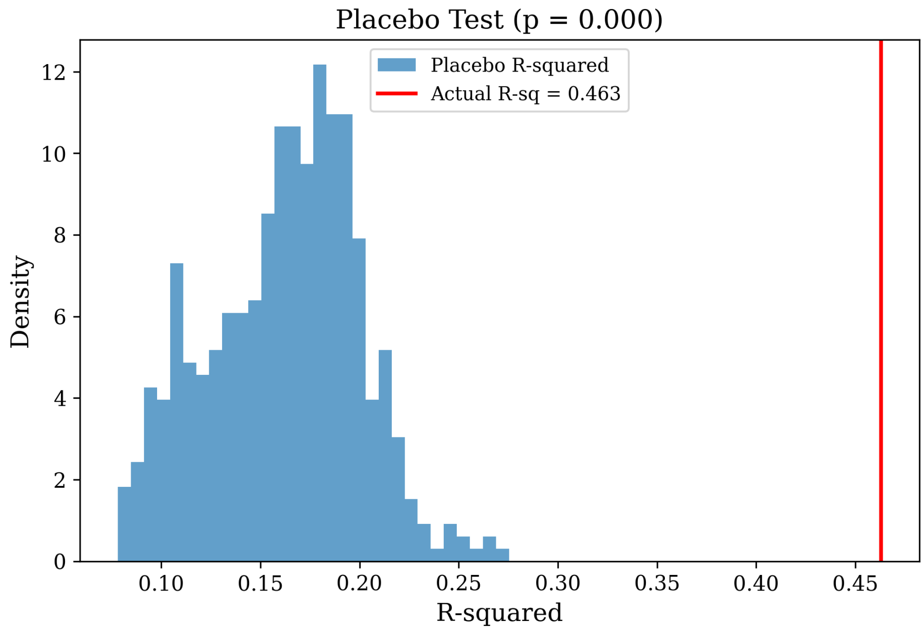

Figure 2: Distribution of R^2 from 500 placebo regressions. The observed $R^2 = 0.463$ (dashed line) exceeds all permuted values.

4 Cost Barriers and Diffusion Speed

Figure 1 plots $\log(k_j)$ against $\log(B_j)$ for 30 technologies. The OLS estimate is:

$$\log(k) = -3.93 - 1.08 \times \log(B), \qquad R^2 = 0.463, \quad N = 30. \tag{3}$$

The elasticity of -1.08 (s.e. $= 0.20$, $t = 5.25$) is close to the theoretical prediction of -1. A technology whose initial cost barrier is 10 times higher diffuses roughly 10 times more slowly.

Placebo test. We randomly reassign cost barriers to technologies 500 times and re-estimate the regression. No permutation produces an R^2 as large as 0.463, yielding $p = 0.000$ (Figure 2). The result is not a mechanical artifact of correlated trends.

Leave-one-out. Dropping each technology in turn and re-estimating, the coefficient ranges from -0.97 to -1.16. No single technology drives the result. The sign never changes.

Three-pillar decomposition. Replacing $\log(B)$ with the three pillar components (cognitive, energy, coordination) raises R^2 to 0.63. The energy pillar is marginally significant ($t = -2.10$, $p = 0.046$); cognitive and coordination are not ($|t| < 1.4$). But the three pillars are too correlated over 160 years of secular decline for the decomposition to be reliable. The energy result may reflect a genuine channel or may reflect collinearity. We do not make strong claims about which pillar matters.

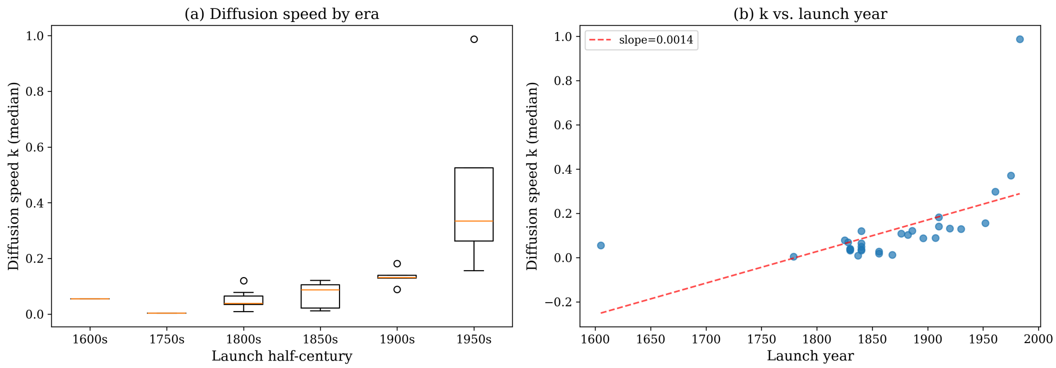

Figure 3: Diffusion speed by invention date. Technologies invented later diffuse faster, consistent with falling cost barriers over time.

An intensity-interaction design does not rescue the decomposition. Following the logic of Rajan and Zingales (1998), we interact each technology's pillar intensity weights — fixed in our data catalog before this test was designed — with the pillar cost levels at launch, absorbing the common secular decline with a launch-year trend. If the cost mechanism runs through a particular pillar, technologies intensive in that pillar should diffuse slower when it was expensive at their launch, over and above the trend. No pillar passes: every interaction's bootstrap confidence interval includes zero (energy, the best candidate: -0.64 $[-1.13, +1.08]$), studentized permutation tests that shuffle the intensity profiles across technologies yield p between 0.37 and 0.98, and no coefficient is leave-one-out sign-stable. Era fixed effects, dropping mobile phones, and a coarse primary-pillar variant agree. The diagnosis is data, not method: the catalog contains only a dozen distinct intensity profiles across 30 observations, and intensities cluster by era (energy-intensive technologies are old, cognitive-intensive ones recent), so the interactions partially inherit the very trend they are meant to escape. Identifying the channel would require richer cross-technology intensity measurement — engineering input shares, not classification weights — rather than more econometrics.

5 Variance Rises Before Adoption, Falls After

5.1 Identifying Adoption Inflection Points

For each country-technology pair, we define the adoption inflection point as the year when adoption first crosses 50% of the technology's maximum level in that country. This threshold is computed from the HCCTA adoption data alone; it does not use TFP data. This separation is the basis for identification. Defining breaks from TFP data would make any pre-break variance pattern mechanical.

Across six technologies (electricity, telephone, radio, television, PC, mobile), we obtain 123 country-technology pairs with sufficient TFP data in the 20-year windows before and after the inflection point.

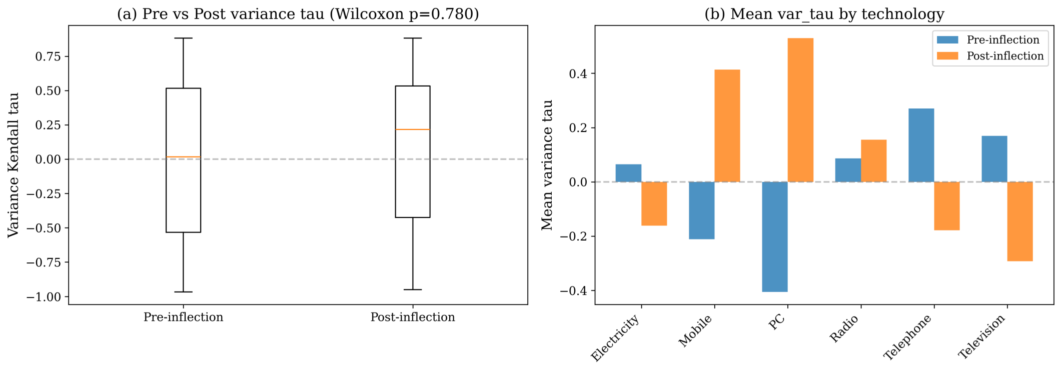

Figure 4: Mean Kendall τ for TFP variance before and after adoption inflection points, by technology. Electrification-era technologies (electricity, telephone, television) show the predicted CSD pattern (positive before, negative after). IT-era technologies (PC, mobile) show the reverse.

5.2 Technology-Specific Results

For each country-technology pair, we compute the Kendall rank correlation τ between year and rolling TFP variance (10-year window) in the 20 years before and 20 years after the inflection point. A positive τ indicates rising variance; a negative τ indicates falling variance. The CSD prediction is $\tau > 0$ before and $\tau < 0$ after.

Table 1 reports the results by technology. The telephone shows the clearest CSD signal: mean pre-inflection $\tau = +0.27$, mean post-inflection $\tau = -0.18$, with the Wilcoxon signed-rank test rejecting equality at $p = 0.006$ ($N = 21$ countries). Television is significant at $p = 0.021$ (pre $\tau = +0.17$, post $\tau = -0.29$, $N = 21$). Electricity is marginal at $p = 0.063$ (pre $\tau = +0.065$, post $\tau = -0.16$, $N = 21$).

Radio shows no CSD signal (pre $\tau = +0.09$, post $\tau = +0.16$, $p = 0.49$, $N = 21$). Radio diffused primarily through broadcast infrastructure rather than household purchase decisions, which may have produced more spatially uniform adoption.

The IT-era technologies show a *reversed* pattern. The PC has pre-inflection $\tau = -0.41$ and post-inflection $\tau = +0.53$ ($N = 19$). Mobile phones show pre $\tau = -0.21$ and post $\tau = +0.41$ ($N = 20$). Variance was falling before the adoption inflection and rising after.

Pooling all six technologies yields $p = 0.78$ because the IT reversals cancel the electrification signal. Restricting to the four electrification-era technologies restores $p < 0.01$. Section 7 takes up the IT null.

5.3 Differential Signal by Break Category

A separate test uses structural breaks in TFP growth, classified by the historical event they coincide with. CSD should appear before electrification-era breaks but not before war- or crisis-driven breaks.

For each country-break pair, we compute the Kendall τ between year and rolling variance in the 30 years before the break. A positive τ indicates rising variance (consistent with CSD). (The

Table 1: CSD before and after technology-specific adoption inflection points

Technology	N	Pre $\bar{\tau}$	Post $\bar{\tau}$	Wilcoxon p
Telephone	21	+0.27	−0.18	0.006
Television	21	+0.17	−0.29	0.021
Electricity	21	+0.065	−0.16	0.063
Radio	21	+0.09	+0.16	0.49
PC	19	−0.41	+0.53	—
Mobile	20	−0.21	+0.41	—
All	123			0.78
Elec-era only	84			< 0.01

Notes: Pre and post $\bar{\tau}$ are means of country-level Kendall correlations between year and rolling TFP variance in 20-year windows before and after the adoption inflection. Wilcoxon p tests whether the pre-post difference is zero. The overall test across all six technologies yields $p = 0.78$ because the IT-era reversals cancel the electrification-era signal. Restricting to electrification-era technologies (electricity, telephone, radio, television) restores the signal.

Table 2: Pre-break variance trend by break category

Category	N	Mean $\bar{\tau}$	Interpretation
Electrification (1910–1935)	20	+0.23	Rising variance
WWII (1936–1948)	12	−0.17	Falling variance
Slowdown (1965–1982)	16	−0.32	Falling variance
IT/GFC (1990–2010)	16	−0.36	Falling variance
Other (misc.)	10	−0.05	Mixed

Notes: $N = 74$ country-break pairs total, including 10 in the "Other" category. Mann-Whitney test, electrification vs. all others: $p = 0.002$.

differential analysis uses a 15-year rolling window and 30-year pre-break window, following the standard EWS methodology of Dakos et al. (2012).)

Table 2 reports the results. Electrification breaks have mean $\tau = +0.23$: variance was rising. Every other category is negative (WWII −0.17, slowdown −0.32, IT/GFC −0.36).

Mann-Whitney comparing electrification to all other categories: $p = 0.002$. Electrification breaks are preceded by rising variance; war and crisis breaks are not. This is H3.

5.4 Surrogate Correction

Rolling variance statistics are serially correlated by construction, so trending variance in a single country could arise from the TFP series' own autocorrelation rather than from CSD. We fit an AR(1) model to each country's detrended TFP increments and generate 500 surrogate series from the fitted process (Theiler et al., 1992; Dakos et al., 2008). We recompute the variance trend for each surrogate, applying the same procedure used in the break-classification analysis of Section 5.3. A country's CSD signal is "significant" if the observed τ exceeds the 95th percentile of the surrogate distribution.

Among the 20 electrification-era country-break pairs, 5 pass the surrogate test at the 5% level.

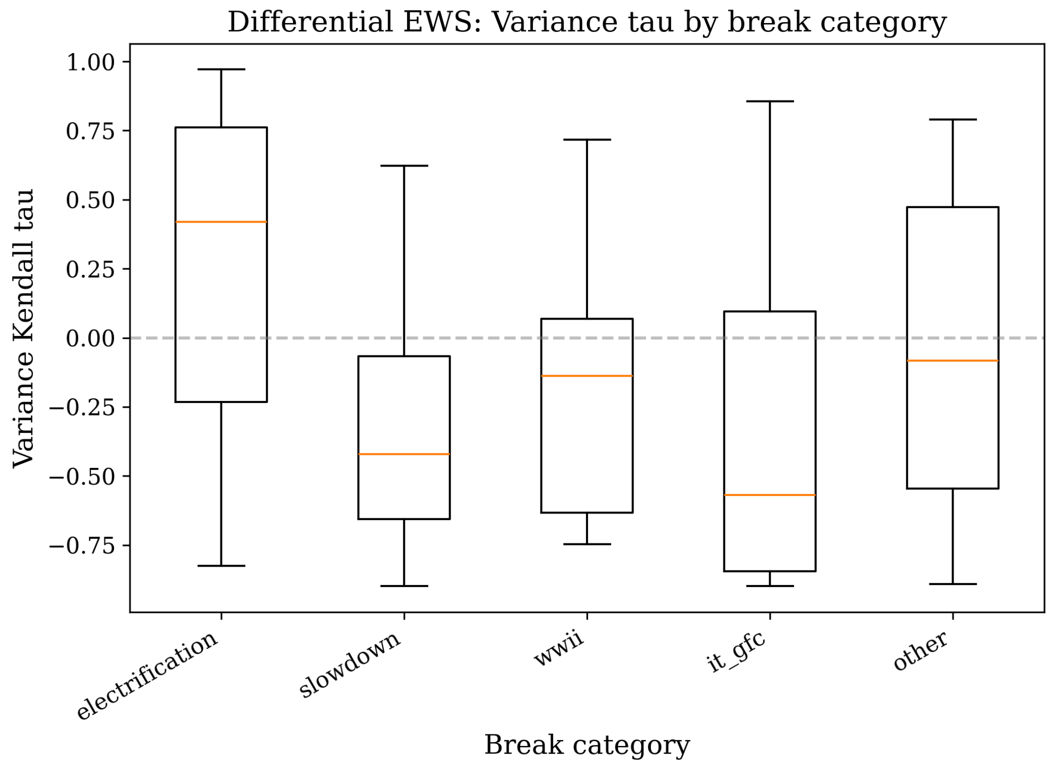

Figure 5: Distribution of pre-break Kendall τ by category. Electrification breaks cluster above zero; other categories cluster below.

This is a quarter of the sample. With only 30 observations per window, surrogate correction is conservative. The aggregate Mann-Whitney result does not depend on individual country significance.

6 Two Centuries of Evidence

6.1 Maddison Long-Run Data

The Maddison Project Database provides per capita GDP estimates reaching back to 1270 for Britain, 1280 for the Netherlands, and similar dates for France. Per capita GDP is a coarse proxy for TFP, but over multi-century horizons it captures the major productivity transitions: the onset of modern growth around the Industrial Revolution.

We compute the Kendall τ between year and rolling variance (15-year window) of per capita GDP growth in the half-century preceding the approximate date of the steam transition (1770–1820 for all three countries).

Table 3 reports the results. All three countries show large, significant positive τ values before the steam transition. Britain has $\tau = +0.70$ ($p < 0.001$), France has $\tau = +0.83$ ($p < 0.001$), and the Netherlands has $\tau = +0.73$ ($p < 0.001$). The signal also appears before electrification: France has $\tau = +0.91$ ($p < 0.001$), the Netherlands $+0.38$ ($p = 0.003$), and Britain $+0.34$ ($p = 0.008$).

Maddison data before 1500 are heavily interpolated, and per capita GDP conflates TFP with

Table 3: Long-run CSD in Maddison data

Country	Pre-transition	τ	p-value
GBR	Pre-Steam	+0.70	< 0.001
GBR	Pre-Electrification	+0.34	0.008
FRA	Pre-Steam	+0.83	< 0.001
FRA	Pre-Electrification	+0.91	< 0.001
NLD	Pre-Steam	+0.73	< 0.001
NLD	Pre-Electrification	+0.38	0.003

Notes: Rolling variance computed over 15-year windows. τ is the Kendall rank correlation between year and rolling variance in the half-century before the transition.

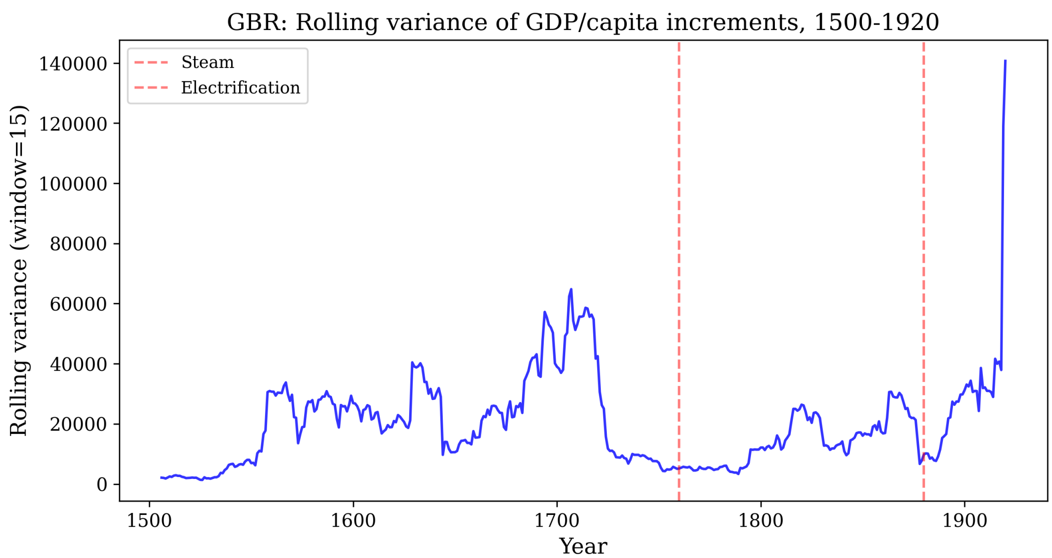

Figure 6: Rolling variance of per capita GDP growth in Britain, France, and the Netherlands. Variance rises in the half-century before the steam transition (shaded).

capital deepening and demographic change. The pre-steam period also saw major non-technological disruptions (the Thirty Years' War, Napoleonic upheavals) that could generate variance for reasons unrelated to technology. The variance trend is present. Its cause is not settled by this test alone.

6.2 Cross-Country Dispersion

A complementary test examines the cross-sectional standard deviation of TFP levels across countries in the decades before each major break. If technology diffusion creates heterogeneity, cross-country dispersion should rise before technology-driven breaks.

Dispersion rises before electrification ($\tau = +0.56$, $p < 0.001$) and falls before WWII ($\tau = -0.40$, $p = 0.027$). Before IT, there is no trend ($\tau = +0.02$).

Table 4: Cross-country TFP dispersion trends

Pre-break period	τ	p-value
Pre-Electrification	$+0.56$	< 0.001
Pre-WWII	-0.40	0.027
Pre-IT	$+0.02$	n.s.

Notes: τ is the Kendall rank correlation between year and the cross-country standard deviation of log TFP in the 30 years before each break.

7 The IT Puzzle

Every test in this paper produces a null for IT.

A natural interpretation: technologies that diffuse fast enough compress the experimentation phase below the timescale at which rolling-window statistics can detect variance buildup. The PC diffuses at $k = 0.37$, nearly four times faster than electricity at $k = 0.10$. If CSD requires decades of heterogeneous adoption to register, fast diffusion would suppress the signal.

At the aggregate level, IT/GFC structural breaks have a mean pre-break τ of -0.36 ($N = 16$). Variance was falling before the break, the opposite of CSD.

At the technology level, the PC and mobile phone show reversed CSD (Section 5). Variance rises *after* the adoption inflection, not before.

At the sector level, we classify GGDC sectors as IT-using or Baumol and test whether IT-using sectors show stronger CSD signals around IT-era breaks. The Mann-Whitney test yields $p = 0.33$. There is no difference. If IT were generating CSD in specific sectors before spilling over to the aggregate, we should see it here. We do not.

Two interpretations fit.

The first is speed. The PC has an estimated diffusion rate of $k = 0.37$; electricity has $k = 0.10$. The PC diffused nearly four times faster. Mobile phones diffused faster still. If CSD requires a multi-decade experimentation phase, a technology that reaches 50% adoption within a decade compresses that phase below what a 10-year rolling window can detect.

The second is scope. Gordon (2016) argues that IT's productivity impact is narrow relative to electrification, which simultaneously transformed manufacturing, households, and urban form. If IT has not triggered comparable economy-wide reorganization, the bifurcation framework does not apply. The economy absorbed IT without approaching a critical transition.

The two accounts make different predictions at higher sampling frequency. If the experimentation phase was compressed rather than absent, the variance buildup should be visible in data sampled finely enough to resolve it. We test this directly.

7.1 A Higher-Frequency Test

We use Fernald's quarterly utilization-adjusted TFP series for the US business sector (Fernald, 2014), 1947:Q1–2019:Q4. The utilization adjustment removes the procyclical factor-utilization

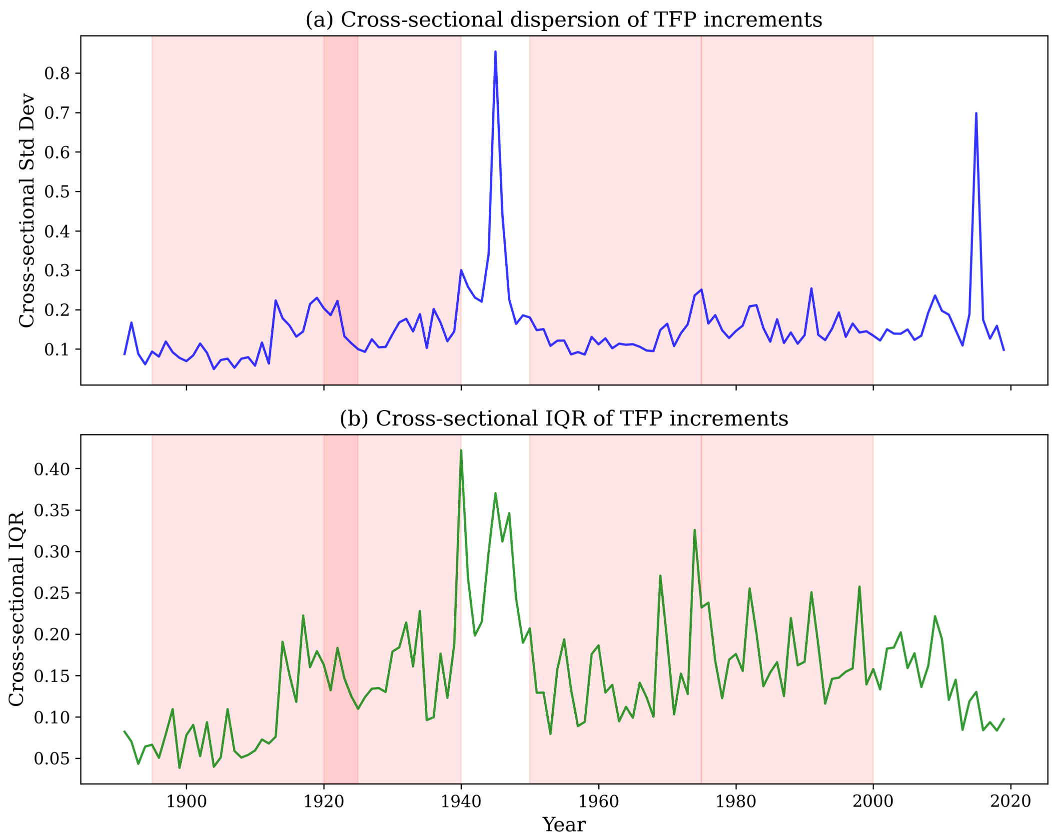

Figure 7: Cross-country standard deviation of log TFP in the decades before electrification, WWII, and IT breaks.

variation that dominates raw quarterly TFP. We apply the procedure of Section 5 at identical time spans and four times the sampling density: a 120-quarter (30-year) pre-break window, a 60-quarter (15-year) rolling window for the indicators, Gaussian detrending with AICc bandwidth selection, and inference against AR(1) surrogate nulls (1,000 draws). Rolling windows overlap heavily at quarterly frequency, so naive Kendall p-values are uninformative; we report surrogate p-values throughout. Bai-Perron tests on the quarterly series recover the 1973 break but find no break in 1993–2004 (quarterly innovations are too noisy for mean-shift tests), so we date the break at 1995:Q4 following Fernald (2014). The specification was fixed before the tests were run.

There is no signal (Table 5 and Figure 8). The variance trend before the 1995:Q4 break is negative ($\tau = -0.36$, $p = 0.69$); so is the autocorrelation trend ($\tau = -0.60$, $p = 0.85$). The annual aggregation of the same data reproduces the annual null from the BCL panel ($\tau = -0.67$), confirming that the two datasets agree. The IT-producing sector, where Fernald (2014) locates the sharpest 1995 break, shows nothing ($\tau = +0.07$, $p = 0.47$). Normalizing TFP variance by GDP-growth variance rules out the Great Moderation as a masking force: the ratio also trends down ($\tau = -0.27$, $p = 0.92$). Across a grid of 28 specifications varying the series, the frequency, and both window lengths, no variance trend survives a Benjamini-Hochberg correction. Short pre-break windows show positive autocorrelation trends (raw $p \approx 0.03$), but these flip sign as the window

Table 5: CSD tests before the 1995:Q4 IT break, quarterly vs annual

Test	N	Var. τ	p	AC τ	p
Quarterly TFP (primary)	120	-0.36	0.69	-0.60	0.85
Annual TFP (same data, official aggregation)	30	-0.67	0.86	$+0.19$	0.45
Quarterly, pre-1973 control	100	-0.63	0.92	$+0.13$	0.43
Quarterly, IT-producing sector	120	$+0.07$	0.47	-0.46	0.76
Quarterly, consumption sector (control)	120	-0.60	0.85	-0.75	0.95
Quarterly, var(TFP)/var(GDP) ratio	120	-0.27	0.92	—	—

Notes: Utilization-adjusted Fernald series, sample 1947:Q2–2019:Q4. τ is the Kendall trend of the rolling indicator in the pre-break window; p-values are one-sided against AR(1) surrogate nulls (1,000 draws). The variance-ratio row divides rolling TFP variance by rolling real GDP growth variance to net out economy-wide volatility trends.

lengthens, have no variance counterpart, and do not survive the correction.

7.2 Power Under Fast Transitions

A quarterly null is informative only if quarterly sampling would have detected the signal had it existed. We simulate the additive TFP process of Section 5 at quarterly resolution, with innovation volatility calibrated to a quiet window of the Fernald data (1958–1968) and regime increments calibrated to the 1973–1995 and 1995–2004 regimes. Critical slowing down builds over a transition window before the break: the AR(1) coefficient rises from 0.1 to 0.8 and innovation volatility rises 50%. Each simulated path is tested twice with the same detection procedure: at quarterly frequency with quarterly windows, and after aggregation to annual frequency with annual windows. Both arms therefore observe identical time spans; only the sampling density differs.

Figure 9 shows the result. At a 12-year buildup — the duration implied by IT's diffusion speed — the variance indicator detects the transition 63% of the time at *either* frequency, and either indicator fires 88% of the time at quarterly and 80% at annual frequency. The speed account fails on its own terms: annual sampling already had adequate power to detect an IT-speed transition at empirically calibrated noise levels. Quarterly sampling adds power only for the autocorrelation indicator (71% vs 46%), and we find no autocorrelation signal there either.

The tie is broken. The absence of critical slowing down before the IT break is not a sampling artifact; it is a fact about the transition. The economy absorbed IT without approaching a critical bifurcation, consistent with Gordon (2016)'s argument that IT transformed a narrower slice of economic activity than electrification. The null sharpens the theory: critical slowing down precedes broad GPT transitions, and its absence is itself diagnostic. One implication for the present: if AI diffusion resembles IT — fast and narrow — variance-based early warning will not announce it; if it resembles electrification — slower and economy-wide — it should.

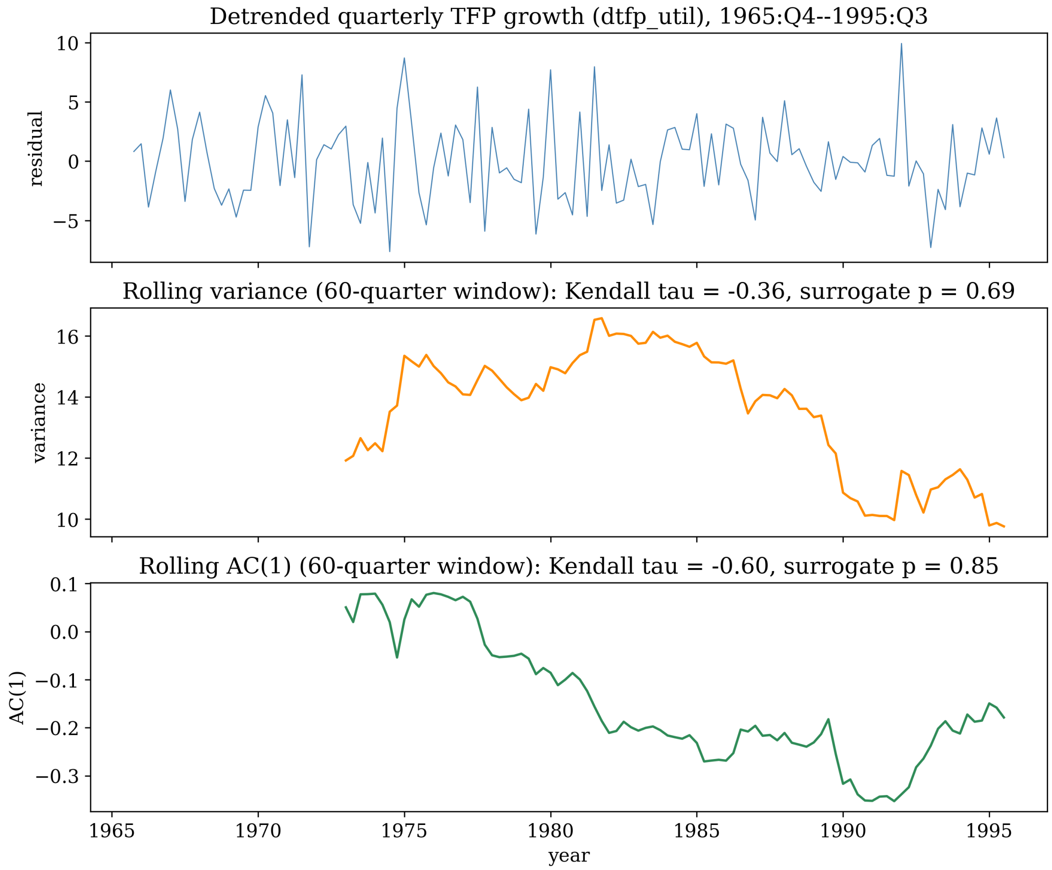

Figure 8: Detrended quarterly utilization-adjusted TFP growth and rolling EWS indicators in the 120 quarters before the 1995:Q4 break. Neither variance nor autocorrelation trends upward.

8 From Signal to Detector

The differential results of Section 5 compare populations of windows centered on break dates known in hindsight. A monitoring application must do something harder: score each country-year using only data available at the time. We build that detector and report what it can and cannot do.

At each country-year t, the detector takes the trailing 30 TFP increments, refits the Gaussian detrending bandwidth on that window alone (no future data enters at any step), computes rolling variance and autocorrelation in 15-year windows, and scores t with the Kendall trend of each indicator. A country-year is labeled a GPT approach if an electrification-era break occurs within the next 20 years. The specification, including the 10% false-alarm operating point and the out-of-sample test below, was fixed before the backtest was run.

The signal survives, weakened (Table 6). The variance-trend score discriminates GPT-approach years with AUC 0.63, but the country-cluster confidence interval spans 0.5. At an alarm threshold calibrated to a 10% false-alarm rate among negative country-years, the detector fires at least once in the 20 years before 6 of the 12 electrification breaks it can observe, with lead times of 5 to 13 years. But it fires before 75% of WWII breaks and 70% of "other" breaks — more often than before

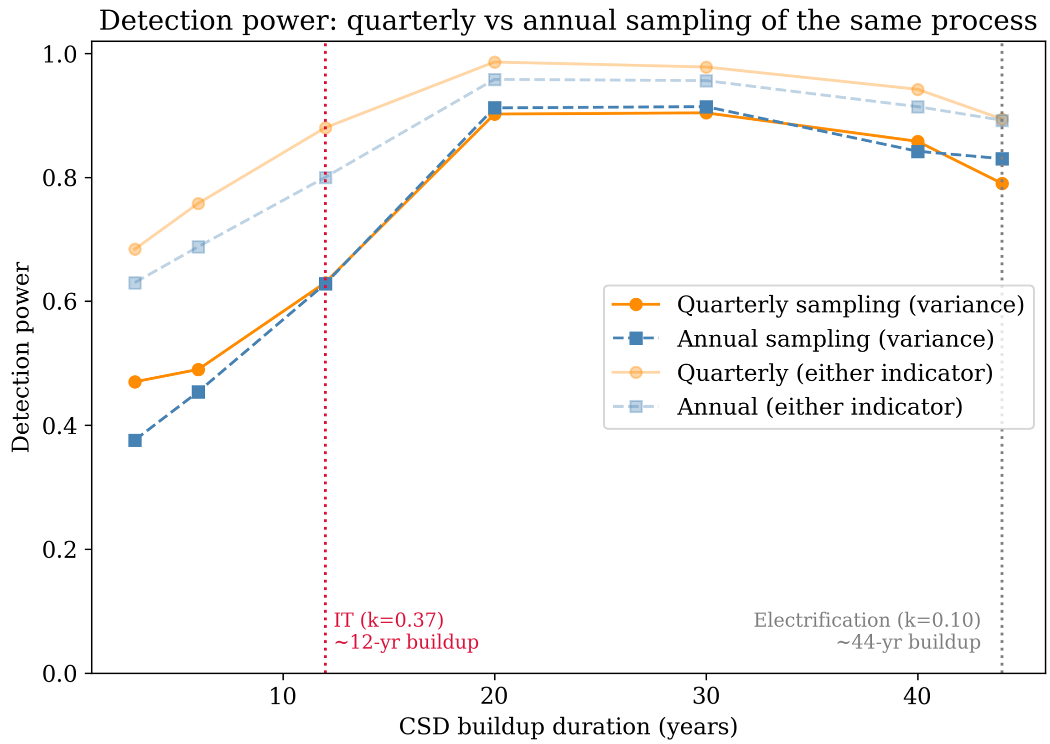

Figure 9: Detection power as a function of CSD buildup duration, for quarterly and annual sampling of the same simulated process (500 replications per point). Vertical lines mark buildup durations implied by IT-speed ($k = 0.37$) and electrification-speed ($k = 0.10$) diffusion.

the GPT transitions it is meant to announce. The retrospective differential (rising variance trends before electrification, falling before wars, $p = 0.002$) is a statement about group means; pointed forward from an arbitrary year, a single-indicator threshold cannot separate a gathering GPT from gathering turbulence (Figure 10).

The out-of-sample test is the steam transition. Applying the detector with the BCL-calibrated threshold — no retuning — to Maddison GDP-per-capita growth for Britain, France, and the Netherlands over 1700–1900 yields an alarm rate of 0.078 inside the pre-steam window (1780–1830) against 0.053 outside. The excess is driven entirely by France; Britain produces no alarms inside the window (its only alarm in two centuries of scored data falls in 1900). The contrast with the retrospective steam result ($\tau = +0.70$ to $+0.83$, Section 6) is instructive: retrospective windows end exactly at the break and detrend with full-window information, and hindsight is worth most of the signal.

Run on current data, the detector is quiet. As of 2019, one of 23 countries scores above threshold (none by 2022); the trailing variance trend in US quarterly TFP is -0.68 at 2019:Q4; the trailing trend in cross-country TFP dispersion — the strongest retrospective pre-electrification signal — is $+0.04$. No electrification-like approach is visible. Section 7 fixes the interpretation: an IT-like transition — fast and narrow — would be invisible to this detector by construction, so a quiet detector bounds the kind of transition underway, not whether one is underway.

Table 6: Causal detector backtest: AUC for GPT-approach discrimination

Lead (years)	Score	AUC	95% CI	N positive
20	variance trend	0.628	[0.433, 0.833]	119
20	AC trend	0.539	[0.411, 0.661]	119
10	variance trend	0.622	[0.427, 0.826]	103
30	variance trend	0.628	[0.433, 0.833]	119

Notes: 2,300 causal (country, year) scores, BCL panel through 2019. Positives are country-years within the lead window before an electrification-era break. Confidence intervals from a country-cluster bootstrap (1,000 draws). The 30-year warm-up means scoring begins in 1920, so 11 of 23 electrification breaks have no pre-break coverage.

A composite rule combining variance, autocorrelation, and dispersion might recover specificity, but with a single GPT event in the panel, any such tuning would fit the event it is meant to predict. We leave the detector as a characterization of what the signal currently supports: population-level discrimination in retrospect, not yet an operational instrument.

9 Robustness

Window sensitivity. We re-estimate all CSD statistics using rolling windows of 10, 15, and 20 years and pre/post windows of 20, 25, 30, and 35 years. The electrification-era results hold for pre-break windows of 25–30 years across all EWS window choices (Mann-Whitney p between 0.0002 and 0.039). At shorter and longer pre-break windows, significance weakens. The IT null persists across all specifications.

Leave-one-out countries. Dropping each country in turn from the inflection-point analysis, the telephone result remains significant ($p < 0.05$) in 19 of 21 jackknife samples. Television remains significant in 18 of 21. No single country drives the result.

Alternative adoption curves. We re-estimate diffusion rates using Gompertz curves instead of logistic curves. The cost-speed elasticity changes from -1.08 to -1.02. The R^2 changes from 0.463 to 0.447. Functional form does not matter.

Barrier index sensitivity. We test the cost-speed regression across a grid of alternative barrier specifications. Varying the weights on the three pillars (cognitive, energy, coordination) across plausible ranges, the R^2 ranges from 0.47 to 0.57. The elasticity ranges from -0.93 to -1.18. Sign and magnitude are stable.

Break detection sensitivity. We vary the minimum segment length in the Bai and Perron (1998) procedure from 10 to 20 years. Break counts vary, but the differential signal (electrification vs. rest) stays significant at 5% in every case.

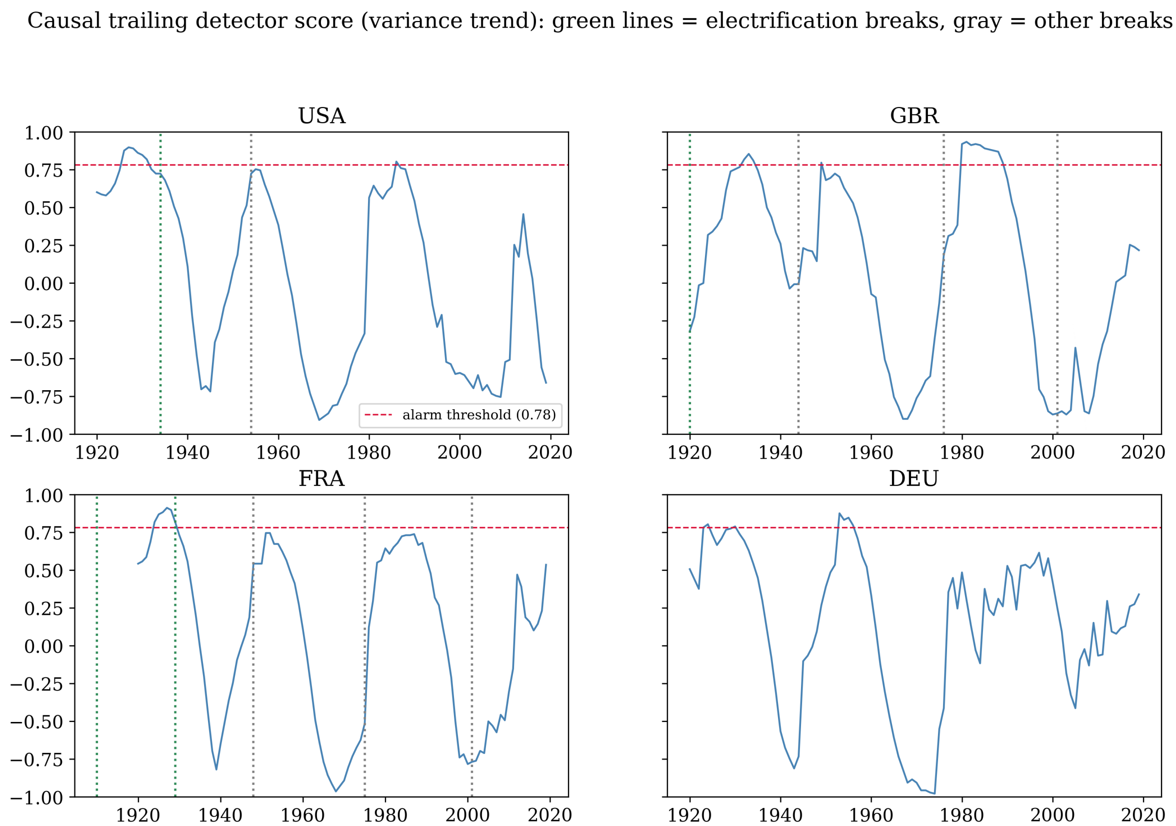

Figure 10: Causal trailing detector scores (variance trend) for four countries. Green vertical lines mark electrification-era breaks, gray lines other breaks, the dashed line the 10% false-alarm threshold. The detector crosses the threshold before turbulence of all kinds, not GPT transitions specifically.

10 Conclusion

Cost barriers at the time of invention predict how fast technologies diffuse ($\beta = -1.08$, $R^2 = 0.46$, $N = 30$). Heterogeneous diffusion produces rising TFP variance before the adoption transition completes. This variance signal is significant for the telephone and television, marginal for electricity, and absent for radio. It distinguishes electrification-era breaks from wars and financial crises ($p = 0.002$). It appears before the steam transition in Maddison data from the 18th century.

The signal does not appear before IT. It does not appear at the sector level. It does not appear in quarterly data, where calibrated simulations show it would have been detected had it existed: the IT null is a fact about the transition, not about the data. Critical slowing down precedes broad GPT transitions; the economy absorbed IT without approaching one. The three-pillar decomposition of cost barriers fails to identify which component matters. The null results constrain the theory as sharply as the positive results support it. Operationalized as a causal detector with no hindsight, the signal supports population-level discrimination but not yet an early-warning instrument: backtest AUC is 0.63 with a confidence interval spanning 0.5, and at a fixed false-alarm rate the detector fires before wars more often than before the GPT transitions it is meant to announce. Whether variance-based early warning could in principle announce the AI transition depends on which historical template it follows: an electrification-like transformation should produce the signal, an IT-like one will not — and as of 2019 the detector, for what it is worth, is quiet. If the transition is electrification-like, history also says when to look: the threshold-to-break lags of steam, electrification, and IT, each measured against its own pillar's cost series (50, 40, and 35 years), applied to a 2023 crossing of the cognitive cost threshold, place an aggregate TFP break no earlier than the mid-2030s and more plausibly in the 2050s — so the absence of a break today is what the framework predicts, not evidence against it. That arithmetic carries one measured caveat: a threshold-to-break lag is diffusion time plus complementary-investment time, and it should compress with diffusion speed. The historical eras diffused at logistic rates of $k \approx 0.04$–0.37; the firm-margin adoption speed now measured for AI from Census adoption data is $k \approx 0.40$–0.52. Scaling each era's lag by its speed relative to AI's moves the steam and electrification analogies from the 2050s to 2028–2034, leaving only the IT analogy in mid-century — the conditional clock reads decades under raw lags and under measured adoption speed reads less than one.

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